

# Case Study - Brickfall Breaker Arcade

A canvas arcade game whose difficulty lies in physics, durable state and coordinated writes

Harbor task: turing/brickfall-breaker-arcade | version 1.0.0 | prepared 12 September 2026

Scope: recorded run outcomes and task description. This document is not an independent fairness assessment or complete QC sign-off. The completed Haiku replacement is included alongside Oracle, GPT-mini and Gemini.

**Recorded scores: Oracle 0.9583 | GPT-5.4 mini 0.6094 | Haiku 4.5 0.1689 | Gemini 3.7 Flash 0.0000 (gate)**

## 1. What the task is

Build a complete web-based brick-breaker from an empty /app. The game should feel responsive immediately and remain correct through seeded checkpoints, a full run, refreshes and concurrent tabs. A convincing canvas alone is insufficient: player identity, game state and results must persist in SQLite.

- Three seeded players - Mira, Dev and Polly - with distinct profiles, checkpoints and history. The demo password is password123.
- Ten workbook-defined levels with normal, strong and solid bricks; fixed 120 Hz simulation; controlled launch/pause/restart; pointer, keyboard, touch and optional Assist.
- Power-ups, combo scoring, extra-life thresholds and terminal bonuses must agree with deterministic fixtures and ordinary gameplay.
- Every state-changing server operation uses an expected revision and unpredictable operation id. Retries return the original receipt; stale tabs reconcile without silently overwriting the winner.
- A non-scoring mechanics lab exposes seven deterministic drills and a visible engine-step counter. Practice must leave ranked saves, best scores, history and leaderboard unchanged.
- Eight supplied instruction files plus workbook/scenario assets define the contract. The agent budget is 7,200 seconds; the separate verifier budget is 13,200 seconds. Both environments use public networking.

## 2. What is verified

Codex / gpt-5.6-luna evaluates the live application using Playwright MCP. The task has 40 criteria in five batched dimensions: 2 Render, 2 Constraints, 25 Functional, 5 Polish and 6 Visual. Functional has 34.5 total criterion weight; the final reward is 60% Functional, 20% Polish and 20% Visual after the Render/Constraints prerequisites.

| Area                     | Examples of graded observations                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Server-backed identity   | Rejected wrong password; issued bearer credentials; protected reads; account revocation and token-owned state despite client identity claims.    |
| Exact state transitions  | Mira/Dev seeded checkpoint outcomes; level progression and terminal records; rising best score retained after reload.                            |
| Mechanics                | Exact drill counters and events; speed/collision handling; multiball, sticky capture, life loss and solid bricks not blocking completion.        |
| Consistency              | Saved-state restoration, retry receipts, stale revisions, distinct operation ids and duplicate-activation suppression.                           |
| Usability and appearance | Semantic HUD/event feedback, keyboard and touch behavior, reduced motion; typography, contrast, layout, hierarchy, craft and mobile consistency. |

### 3. Where each run landed

| Run                     | Overall | Render | Constr. | Func.  | Polish | Visual | Recorded status            |
|-------------------------|---------|--------|---------|--------|--------|--------|----------------------------|
| Oracle                  | 0.9583  | 1.0000 | 1.0000  | 1.0000 | 1.0000 | 0.7917 | 25/25 Functional           |
| GPT-5.4 mini (high)     | 0.6094  | 1.0000 | 1.0000  | 0.3768 | 1.0000 | 0.9167 | 9/25 Functional            |
| Gemini 3.7 Flash (high) | 0.0000  | 1.0000 | 0.0000  | 0.3986 | 0.2000 | 0.9583 | 9/25 Functional; gate-zero |
| Claude Haiku 4.5        | 0.1689  | 1.0000 | 1.0000  | 0.0870 | 0.0000 | 0.5833 | 2/25 Functional            |
| No-op control           | 0.0000  | 0      | 0       | 0      | 0      | 0      | graded=0; no_op=1          |

#### Oracle - 0.9583; Functional 1.0000

The current golden solution passed every one of the 25 Functional criteria. Render, Constraints and Polish also scored 1.0. All 40 criteria are present in the export; no Functional session was omitted. This includes the added checks for higher best-score persistence, forged identity and completing a level with an intact solid brick.

Visual was 0.7917: five axes received 4/5 and mobile consistency received 5/5. The judge described small/dense telemetry, minor muted contrast and lower-page empty space. These are the only recorded Oracle deductions; this run is not an overall 1.0000.

#### GPT-5.4 mini, high reasoning - 0.6094

The model built an application that cleared both prerequisites and scored 1.0 Polish and 0.9167 Visual. Functional was 0.3768: 9/25 binary criteria passed, earning 13 of 34.5 weight. Its recorded overall score falls inside the documented 0.1-0.7 keeper band.

The judge reported a non-finite canvas-gradient error when resuming seeded checkpoints, preventing the terminal/progression journey from completing. Several lab checks observed one or two steps where the required outcomes called for 120 or 240. Other deductions involved manifest digest visibility, complete history snapshots and coordination/security evidence. The evaluation report reproduces the full criterion-level verdicts and explanations.

#### Gemini 3.7 Flash, high reasoning - recorded gate-zero

Render scored 1.0, Functional 0.3986 (9/25), Polish 0.2 and Visual 0.9583. Constraints was 0.0, so the final runner wrote 0.0000. The Constraints judge reported that successful sign-in automatically issued a Start request, and stopped at its non-mutating gate rule. These are the recorded outcomes; the zero is not described as 40 independently demonstrated product failures.

The Functional and Polish explanations repeatedly report createDrillState is not defined during Load Drill. Inspection of the exported source found both physics.js and drills.js declaring the same top-level CONSTANTS in classic browser scripts; a local combined-script syntax check reports the duplicate declaration. The judge also reported that Next Level reset score/lives rather than preserving progression. The run achieved strong Visual ratings despite those recorded functional gaps.

#### Claude Haiku 4.5 - completed at 0.1689

The latest run-8e4c12dd / 6uqBDww completed without a trial exception and cleared both gates. Functional was 0.0870: 2/25 criteria passed, earning 3 of 34.5 weight. Polish was 0.0000 and Visual was 0.5833, giving an overall 0.1689. The previous completed run-05fd6dc8 also scored 0.1689 but had different criterion outcomes; its export and prior delivery remain preserved. The latest run is selected, not the earlier startup-timeout attempt.

Sign-in/token basics and forged-client-identity handling passed. The judge found missing Mira/Dev checkpoints, only four leaderboard rows and failed brick-data requests. Lab Load/Advance controls exist, but telemetry stays blank and the counter stays zero. Exported source corroborates the omissions: no checkpoint import, guest inserts missing a required run\_id, an invalid unquoted SQL alias, unwired Advance controls and a placeholder advance routine. These prevent the exact checkpoint, terminal and drill journeys from being demonstrated.

Important interpretation limit: missing checkpoints cause several dependent failures. The combined session criterion also failed although the judge confirmed distinct tokens and successful account-wide revocation; its remaining objections were missing Dev state and stale resume UI. Do not count each failed criterion as a separate defect or call this a proven logout failure. Source/isolated SQLite checks corroborate the main gaps; this was not a new full browser regrade.

## No-op control - 0.0000

The Oracle job contains a separate empty-submission control. It scored 0.0000 with graded=0 and no\_op=1, distinct from an application that was built and received criterion grades.

## 4. What the observed results show

The strongest concrete result is the separation between the golden implementation and GPT-mini on exact Functional behavior: 25/25 versus 9/25, while both received full Polish credit. Their overall gap is 0.3489. This illustrates why the task checks deterministic state and persistence alongside presentation instead of relying solely on an attractive canvas.

Gemini and GPT each passed nine Functional criteria but earned different weighted scores. Gemini's overall was then gated to zero, so these results do not support a simple product-quality ranking based only on their final totals. Haiku's replacement adds a lower measured result at 0.1689, with both gates passed and two Functional passes.

## 5. Delivery status and evidence

- The reports summarize the supplied platform exports; preparation did not change the task, golden solution, criteria, weights or scores and did not launch a paid run.
- The completed exports share one uploaded task checksum. Current verifier file hashes and golden source match the Oracle evidence; the task ZIP has 32 source-matching files in one correct wrapper.
- Oracle Functional is complete at 1.0000. Overall Oracle is 0.9583, not 1.0000. GPT-mini is 0.6094 and the completed Haiku replacement is 0.1689. All requested exports are included.
- Two completed Haiku trials share the 0.1689 total with different criterion outcomes; the latest is delivered. One selected trial is reported for each other evaluated model. This is not a repeatability study, complete QC sign-off or blanket fairness certification.

Task ZIP SHA-256: e41998a64346e553627ab7ce9263c129c4dcaad66bec8c45aaff71ab887c8c1a

The companion EVAL-REPORT-brickfall-breaker-arcade.docx contains all 40 recorded verifier results, exported failure explanations, score arithmetic and the source/run index. Four named job-directory ZIPs preserve Oracle/no-op, GPT-mini, Gemini and the completed Haiku replacement.